

Properties of Sufficient Statistics

- We will discuss how sufficiency is used to find better estimators.
- Sufficient statistic is not UNIQUE. If $Y_1 = u_1(X_1, \dots, X_n)$ is a sufficient statistic and $Y_2 = g(Y_1)$ where $g(x)$ is a one-to-one function is a statistic then

$$\begin{aligned}f(x_1, \theta) \cdots f(x_n; \theta) &= k_1[u_1(y_1; \theta)]k_2(x_1, \dots, x_n) \\&= k_1[u_1(g^{-1}(y_2); \theta)]k_2(x_1, \dots, x_n),\end{aligned}$$

hence, Y_2 is also sufficient.

- Two old results from Probability:

$$\mathbb{E}[X_2] = \mathbb{E}[\mathbb{E}(X_2|X_1)]$$

$$\text{var}(X_2) \geq \text{var}[\mathbb{E}(X_2|X_1)]$$

- Let Y_1 be the sufficient statistic and Y_2 be an unbiased estimator for θ .
With $\mathbb{E}(Y_2|y_1) = \psi(y_1)$,

$$\theta = \mathbb{E}(Y_2) = \mathbb{E}[\mathbb{E}(Y_2|Y_1)] = \mathbb{E}[\psi(Y_1)]$$

$$\text{var}(Y_2) \geq \text{var}[\psi(Y_1)]$$

$\psi(Y_1)$ is unbiased and have smaller variance than that of Y_2 .

- Theorem: Let $X_1, \dots, X_n$ denote a random sample from pdf or pmf $f(x; \theta), \theta \in \Omega$. Let $Y_1 = u_1(X_1, \dots, X_n)$ be a sufficient statistic for θ , and let $Y_2 = u_2(X_1, \dots, X_n)$, not a function of Y_1 , be an unbiased estimator of θ . Then $\mathbb{E}(Y_2|y_1) = \psi(y_1)$ defines a statistic $\phi(Y_1)$. This statistic $\psi(Y_1)$ is a function of the sufficient statistic for θ ; it is an unbiased estimator of θ ; and its variance is less than that of Y_2 .
- We can restrict our search for a best estimator to function of Y_1 .

- There is a connection between sufficient statistics and maximum likelihood estimates.
- If a sufficient statistic $Y_1 = u_1(X_1, \dots, X_n)$ for θ exists and if a maximum likelihood estimator $\hat{\theta}$ of θ also exist uniquely, then $\hat{\theta}$ is a function of $Y_1 = u_1(X_1, \dots, X_n)$.

- Something is wrong here: $\psi(Y_1) = \mathbb{E}(Y_2|Y_1)$ is unbiased and has smaller variance than Y_2 of θ . Let $\Upsilon(y_3) = \mathbb{E}[\psi(Y_1)|Y_3 = y_3]$. By Rao-Blackwell, $\mathbb{E}(\Upsilon(Y_3)) = \theta$ and has smaller variance than does $\psi(Y_1)$.

- Let X_1, X_2, X_3 be a random sample from an exponential distribution with mean $\theta > 0$. The joint pdf is

$$\left(\frac{1}{\theta}\right)^3 e^{-(x_1+x_2+x_3)/\theta}.$$

$Y_1 = X_1 + X_2 + X_3$ is sufficient, and $\mathbb{E}(Y_1) = 3\theta$. So $\bar{X}$ is a sufficient statistic for θ and unbiased.

Let $Y_2 = X_1 + X_2$ and $Y_3 = X_3$. Then

$$g(y_1, y_2, y_3; \theta) = \left(\frac{1}{\theta}\right)^3 e^{-y_1/\theta}, 0 < y_3 < y_2 < y_1 < \infty.$$

$$g_{13}(y_1, y_3; \theta) = \left(\frac{1}{\theta}\right)^3 (y_1 - y_3) e^{-y_1/\theta}, 0 < y_3 < y_1 < \infty$$

$$g_3(y_3; \theta) = \frac{1}{\theta} e^{-y_3/\theta}, 0 < y_3 < \infty$$

The conditional pdf of Y_1 given $Y_3 = y_3$ is

$$g_{1|3}(y_1|y_3) = \left(\frac{1}{\theta}\right)^2 (y_1 - y_3) e^{-(y_1 - y_3)/\theta}, 0 < y_3 < y_1 < \infty.$$

Thus

$$\mathbb{E}\left(\frac{Y_1}{3} | y_3\right) = \frac{1}{3} \frac{\Gamma(3)\theta^3}{\theta^2} + \frac{y_3}{3} = \frac{2\theta}{3} + \frac{y_3}{3} = \Upsilon(y_3).$$